

M1 MLDM Summer Research Project Report: County-Level Corn Yield Prediction via Meteorological and Soil Sequences

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Abstract

Forecasting crop yields at a regional level is essential for food security and commodity planning. However, predicting corn yields county-by-county is tricky due to unpredictable weather patterns and complex seasonal dynamics. For this project, we built a machine learning pipeline to forecast county-level corn yields across 778 counties. The model uses 30 weather features broken down across five key crop growth stages, combined with baseline soil properties from the USDA SSURGO database (such as available water capacity, clay, sand, and organic matter). To mimic real-world forecasting conditions and avoid data leakage, we set up a strict chronological split—training on data from 2005 to 2022 and testing on 2023 to 2025—alongside time-series cross-validation. We then benchmarked traditional tree-based models and deep learning networks against a naive trend-drift benchmark and sequence-based architectures designed to capture historical weather trajectories. The results show that combining multi-year weather memory with baseline soil characteristics significantly improves prediction accuracy across unseen test years.

1 Introduction

1.1 Problem Statement

Accurately predicting crop yields is a cornerstone of global food security, regional economic planning, and modern agriculture. With climate variability intensifying and extreme weather events growing more frequent, agricultural stakeholders—from local policymakers to commercial supply chain managers—rely on early-stage yield forecasts to anticipate supply shocks, guide grain import/export policies, and optimize resource allocation. However, forecasting crop yields county-by-county presents a major data science challenge due to complex, messy interactions between soil properties, local farming practices, and changing weather patterns. Historically, crop assessments relied heavily on basic statistical models, linear regressions, or gut feeling. While useful for broad historical averages, these conventional methods often miss how weather fluctuations punch holes in harvests—especially during critical moments like pollination—or how local soil conditions either buffer or worsen the damage. To get around these limits, people are turning more and more to machine learning and deep learning.

1.2 Context of the Project

This personal project is motivated by a professional aspiration to join Louis Dreyfus Company, one of the four major agricultural commodity trading houses globally. For such an organization, accurate projection of bushel yields across global production regions is vital for optimizing trade execution, supply chain management, and risk mitigation. This research serves as an applied M1 Master’s project within the Machine Learning and Data Mining (MLDM) program at Université Jean Monnet, intended to showcase rigorous predictive modeling capabilities for professional portfolios in M2 internship applications.

1.3 Survey of the State of the Art

Predicting agricultural crop yield accurately is a critical component of regional food security, economic planning, and modern precision agriculture. Historically, crop assessment relied heavily on straightforward linear math or local intuition, which doesn’t hold up well against complex environmental changes. To fix this, recent research leans heavily on machine learning and deep learning. In a comprehensive literature review by van Klompenburg et al. (2020), the authors looked at dozens of studies to see what features and metrics people use most when forecasting crops. They found that temperature, rainfall, and soil makeup are the absolute go-to inputs. They also noted that standard metrics like the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) are the standard ways everyone checks if their model actually works.

While older machine learning tools—like Random Forests, Support Vector Machines, and basic neural networks—started things off, the field has leaned more toward deep learning lately. Specifically, van Klompenburg et al. (2020) pointed out that sequence-based deep learning models like **Long Short-Term Memory (LSTM)** networks and Convolutional Neural Networks (CNNs) are among the best choices for handling spatial-temporal and weather-driven farm data. Unlike older models that just look at a single year in a vacuum, sequence models catch how weather piles up over time and plays out across multiple years. Building on what others have done, this project pits traditional tree-based models against deep learning sequence networks, using both multi-year weather history and soil profiles.

2 Analysis

2.1 Data Exploration

The dataset comprises 13,736 total observations across 778 unique counties spanning from 2005 to 2025. Each record merges historical NASS corn yield targets with weather data and soil properties. As illustrated in Figure 1, the target variable (corn yield in bushels per acre) follows a fairly normal bell curve centered around 160–175 bu/acre, bounded by natural performance extremes.

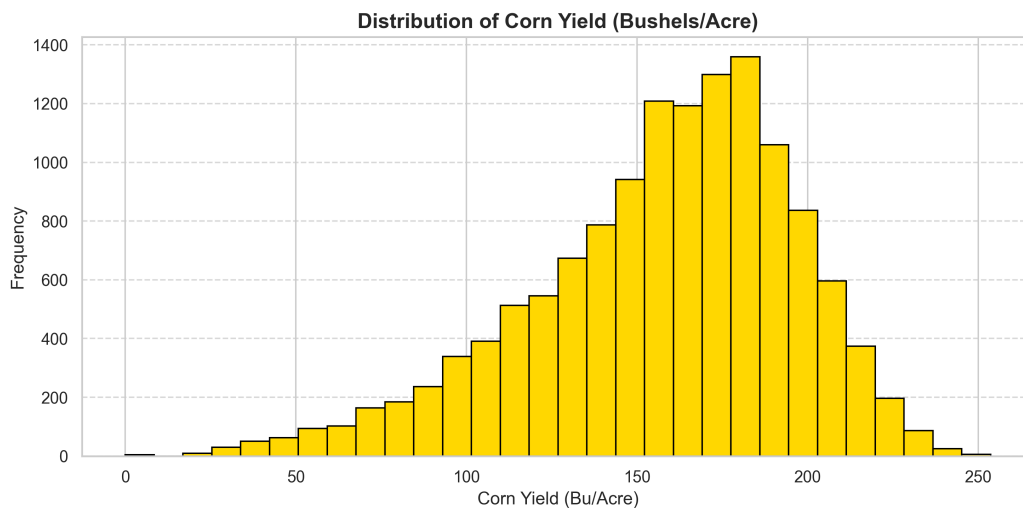


Figure 1: Distribution of county-level corn yields across the dataset.

To understand how county-level yields stack up geographically, Figure 2 displays the overall average corn yield by state across the study period.

If you aren’t acquainted with U.S. farming geography, the chart in Figure 2 clearly points out the famous “Corn Belt”—specifically states like Iowa (IA) and Illinois (IL), which top the national average at close to 180 bushels/acre. This massive advantage comes down to deep, nutrient-rich mollisol soils left behind by ancient glaciers, great summer rainfall, and top-tier farming tech. On the flip side, western and northern

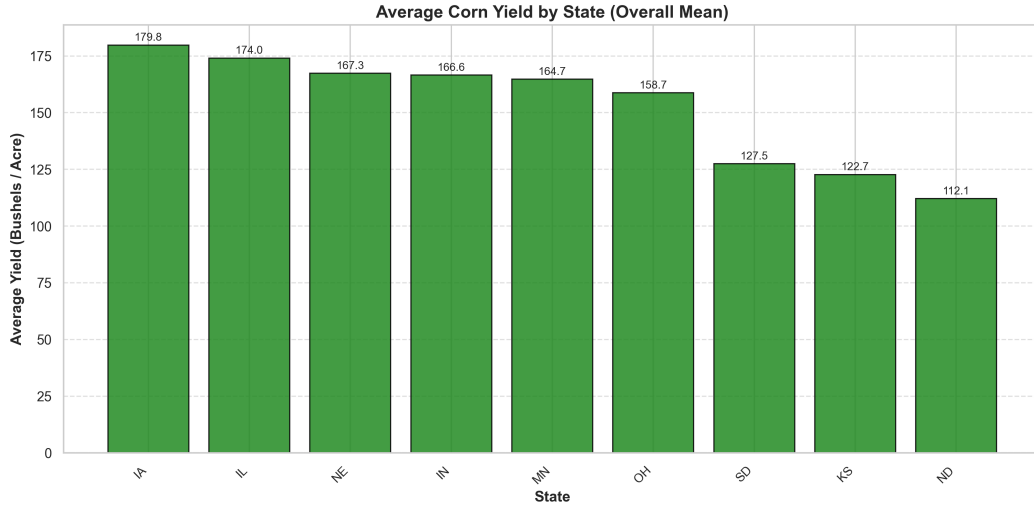


Figure 2: Average corn yield performance aggregated by state.

states like Kansas and North Dakota average lower numbers because they deal with drier climates, shorter growing seasons, and rely a lot more on irrigation or dryland farming limits.

2.2 Exploratory Visualization

To uncover underlying spatial and temporal patterns, exploratory visualizations were mapped across states and historical timelines. Figure 3 highlights macro-level volatility, most notably the severe 2012 drought crash followed by steady modern recovery.

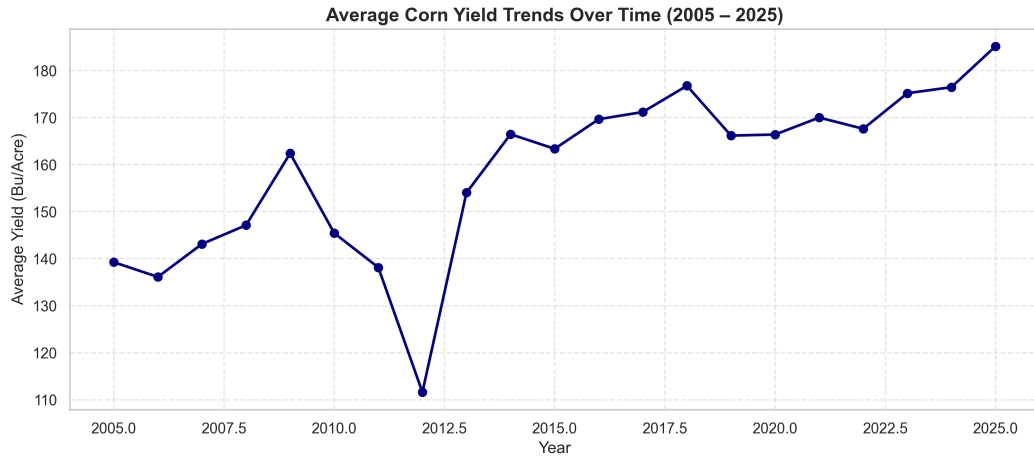


Figure 3: Average national/regional corn yield trends over time (2005–2025).

The national yield path shown in Figure 3 maps out three distinct eras in American farming over these two decades:

1. **The Pre-Drought Era (2005–2011):** A steady period where major farming states hovered between 140 and 160 bushels/acre under normal weather conditions.
2. **The Extreme Drought Shock Year (2012):** The year 2012 was easily one of the worst droughts in modern U.S. history. Brutal heatwaves and almost zero rain right when corn was trying to pollinate

smashed yields across the Midwest. As you can see in the steep drop in Figure 3, the national average tanked to around 112 bushels/acre—way below normal expectations.

3. **The Post-Drought / Modern Era (2013–2025):** After the 2012 wake-up call, farming bounced back fast with better tech. Thanks to drought-hardy seeds, smart irrigation, and precision fertilizer, yields bounced back hard and climbed steadily toward 185 bushels/acre by 2025.

Also, looking at state differences (Figure 2) and soil properties like water capacity and organic matter shows that good soil acts like a shock absorber when the weather turns ugly.

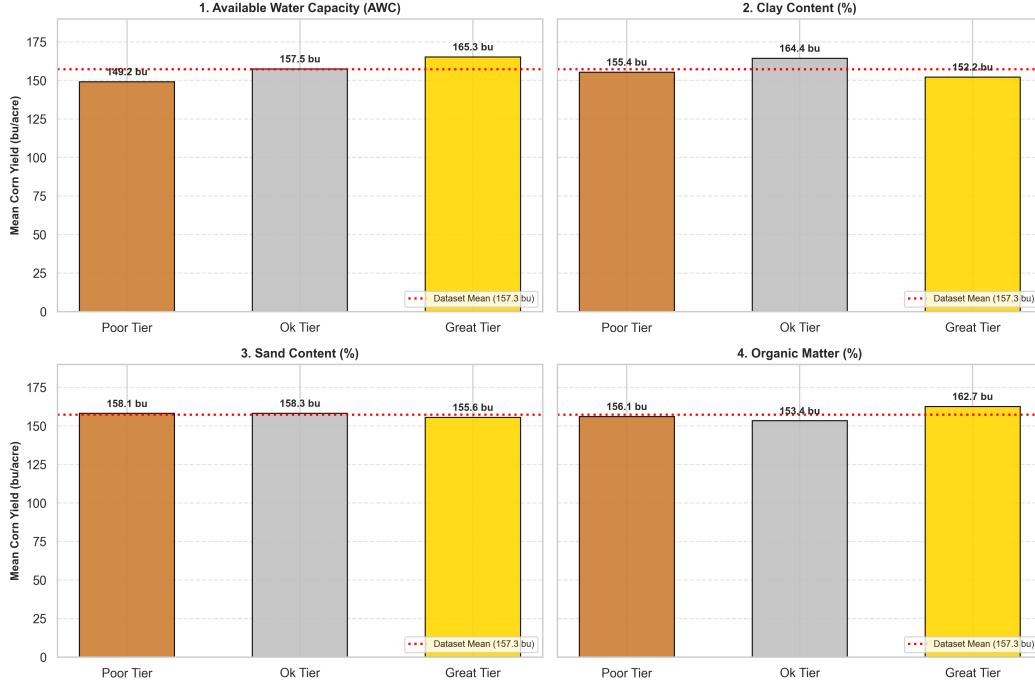


Figure 4: County-level corn yield performance across bronze, silver, and gold tertile tiers of USDA SSURGO soil properties. Higher Available Water Capacity (AWC) and Organic Matter (OM) directly correspond to increased bushel yields in the top (gold) tier, outperforming the national dataset mean benchmark.

3 Algorithms and Techniques

To capture both non-linear environmental thresholds and time-based trends, a mix of algorithms was put to work:

- **Tree-Based Ensembles:** Random Forest, XGBoost, and LightGBM regressors, picked because they handle messy tabular feature mixes and sharp cutoffs really well.
- **Stacking Regressors:** A meta-model that combines tuned tree models with a regularized Ridge Regression to smooth out individual blind spots.
- **Sequence-Based Neural Networks:** Custom PyTorch implementations of Long Short-Term Memory (LSTM) networks and 1D Convolutional Neural Networks (CNNs) built to digest multi-year weather sequences ($seq_length = 3$).

4 Benchmark

To prove whether fancy machine learning models actually add value over simple historical momentum, we set up a **Naive Trend-Drift Benchmark**. In agriculture, past yields naturally lean on long-term trends like better tech, stronger seeds, and improved farming habits.

As literature points out, comparing models against a trend baseline is a must because crop data naturally marches upward over time. As van Klompenburg et al. (2020) noted in their review, evaluation setups have to account for underlying trends so that models actually “*assess model generalizability and predictive accuracy*” rather than just blindly copying past growth. Using a drift baseline ($\hat{y}_t = y_{t-1} + \Delta_{\text{avg}}$) forces our machine learning models to clear a real hurdle—proving they catch actual mid-season weather shocks, heat stress, and soil-climate quirks instead of just coasting on historical momentum. Beating this baseline proves the models are actually useful for real-world commodity forecasting.

5 Methodology

5.1 Data Preprocessing

Missing values across weather and soil columns were filled in using mean substitution. To keep things clean and avoid data leakage, feature scaling ($\mu = 0, \sigma = 1$) for the neural networks was done using math computed entirely from the development window (2005–2022). We also split the data strictly by time:

- **Development Period (2005–2022):** Made up of 12,083 rows, used for exploring the data, cross-validation, tuning, and initial training.
- **Test Period (2023–2025):** Made up of 1,653 rows across 696 counties, locked away and untouched until final testing.

5.2 Feature Engineering

Raw weather and soil numbers on their own don’t capture how plants actually experience stress. To fix that, we built features around five major growth stages of corn:

1. **Recharge:** The pre-season moisture buildup that sets up underground water storage.
2. **Planting:** The early sprouting and root growth phase, which cares about early heat and moisture.
3. **Pollination:** The critical reproductive window when silking happens; highly sensitive to extreme heat and dry air.
4. **Grain Fill:** The period where kernels pack in starch and weight, deciding the final harvest weight.
5. **Harvest:** The dry-down and wrap-up stage before pulling the crop out of the field.

To give the models better signals, we engineered several new features:

- **Interactive Stress & Aridity Indices:** Multiplying vapor pressure deficit and max temperature during touchy stages like pollination ($\text{vpd_pollination} \times \text{tmax_pollination}$) and making aridity ratios ($\text{precip_pollination} / (\text{pet_pollination} + 1e - 5)$).
- **Soil-Climate Interactions:** Multiplying soil water capacity by grain-fill vapor pressure deficit ($\text{MEAN_AWC} \times \text{vpd_grain_fill}$) to see how well moisture-holding soils block heat stress.
- **Historical Lag Features:** Adding `yield_lag_1`, `yield_lag_2`, and `yield_lag_3` to track multi-year trends and local farming momentum.

Figure 5 shows the dual-axis trend comparing average rain during planting and pollination against the national corn yield trend over time, showing how seasons tie together.

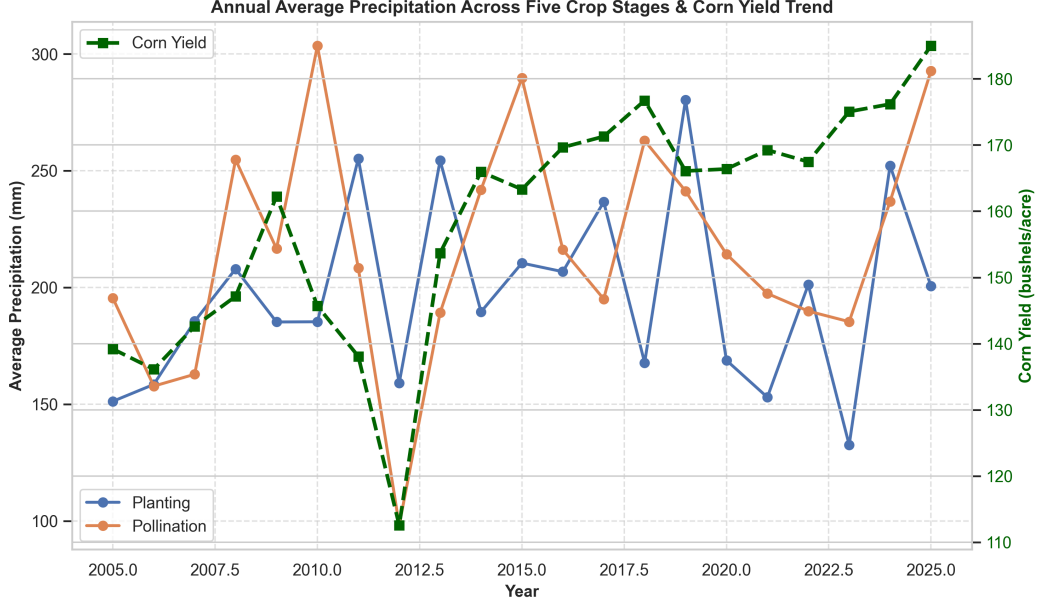


Figure 5: Annual average precipitation across key crop stages alongside national corn yield trends.

5.3 Leakage-Free Z-Score Weather Anomalies

To help tree models and neural nets make sense of weather across hundreds of completely different counties without getting confused by local climates, we turned the weather data into county-specific Z-score anomalies.

Raw weather stats can easily trip up a model. Imagine getting 60 mm of rain: in a humid farming county in North Carolina, 60 mm during a key window spells total drought disaster. But that exact same 60 mm might be completely normal for a drier county in North Dakota. If a model only looks at raw rain totals, it won't know the difference between a crisis and a normal Tuesday.

To clear this up, anomaly math answers a simple question: *“How weird was the weather here compared to what’s normal for this spot?”* Every weather feature was measured against that specific county’s own history, calculated strictly from the development years (2005–2022):

$$\text{Anomaly} = \frac{\text{Raw Value} - \text{County Historical Mean}}{\text{County Historical Standard Deviation}}$$

By turning raw numbers into standard deviations away from the norm, every county gets put on the same statistical playing field. A 0.0 means completely normal weather, while a -2.0 screams severe drought, whether it’s happening in Iowa or Kansas. This strips out local background noise so models like XGBoost and LightGBM can focus purely on the unusual weather swings that actually cause crop stress.

5.4 Feature Importance and Soil Significance

To figure out which environmental and farming factors matter most, we pulled feature importance scores from the tuned Random Forest model (Figure 6).

As Figure 6 shows, past yield lags (`yield_lag_1`, `yield_lag_3`) score super high because they implicitly capture local management habits, seed upgrades, and baseline soil fertility. But among active weather variables, **aridity_pollination** pops up as the second most important feature overall.

To see how different seasons stack up, we grouped feature importances by crop growth stage (Figure 7). The breakdown proves that weather during the **pollination** stage alone drives over 45% of the model’s total predictive power, with **grain fill** chipping in another 21%.

Why pollination ranks so high comes down to corn biology. In the middle of summer, silking and pollination make or break the harvest. If the air gets too hot and dry, pollen dies off and silks dry out

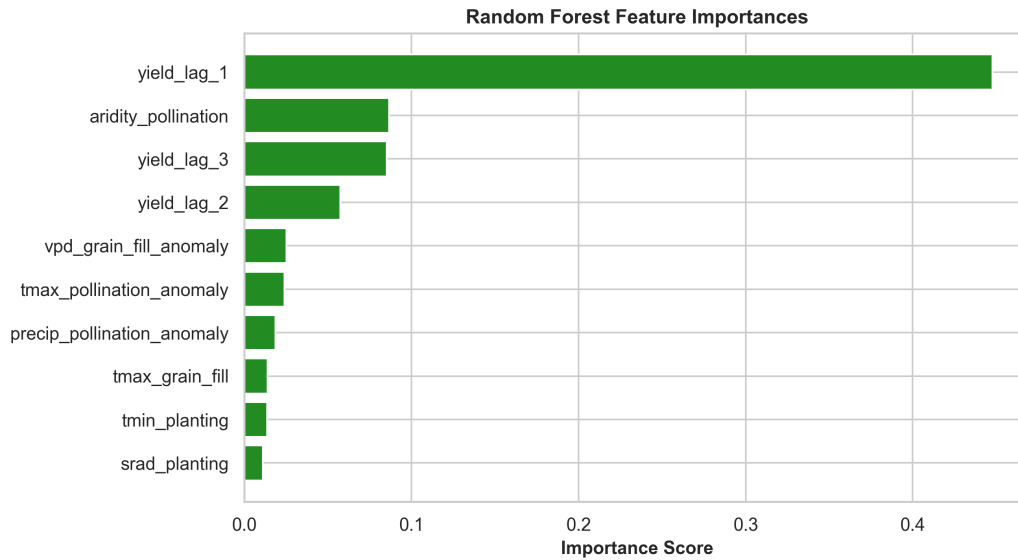


Figure 6: Random Forest feature importance rankings highlighting the dominance of historical lags, aridity indices, and soil-weather interactions.

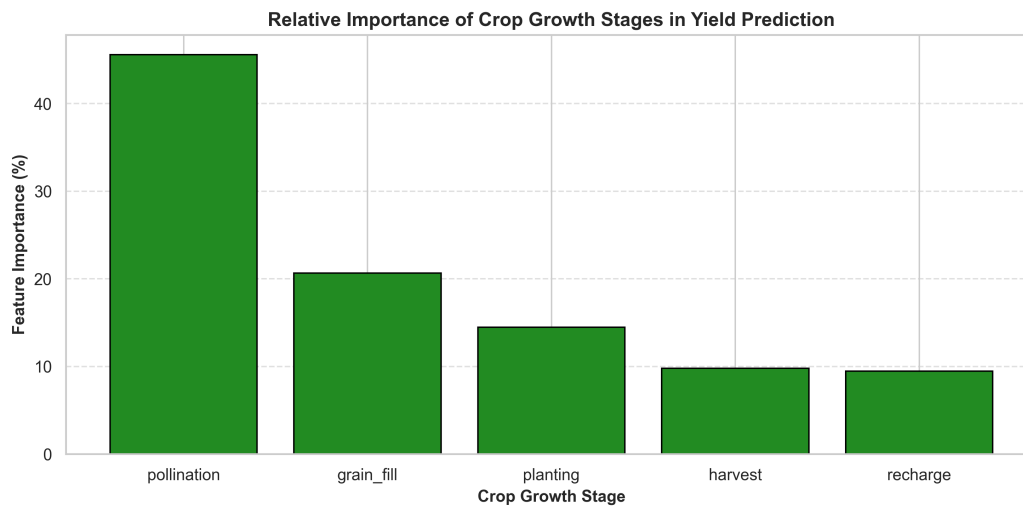


Figure 7: Aggregated feature importance percentage by crop growth stage, demonstrating the overwhelming dominance of the pollination and grain fill windows.

before they can catch anything, leading to massive kernel loss no matter how great the spring weather was. Mathematically, `aridity_pollination` looks at the ratio of rain to Potential Evapotranspiration ($\text{precip_pollination}/(\text{pet_pollination} + 1e - 5)$). While raw rain numbers can lie during a heatwave because water evaporates instantly, this aridity ratio nails the balance between water supply and atmospheric thirst. Tree models love this metric because it cleanly shows whether crops hit trouble during their most vulnerable window.

On top of that, adding soil-weather interactions and anomalies helped the models catch local drought defense. Soil traits like Available Water Capacity (`MEAN_AWC`) and Organic Matter (`MEAN_OM`) act like built-in sponges: counties with higher AWC stay stable during dry years because their soils hold onto moisture right through the toughest heat spells.

5.5 Implementation

Feature engineering included building stress indices (like $\text{VPD} \times \text{Max Temp}$), aridity ratios, soil-climate interaction terms ($\text{MEAN_AWC} \times \text{VPD_grain_fill}$), leak-free Z-score anomalies based on training data, and historical lags (`yield_lag_1`, `yield_lag_2`, `yield_lag_3`). Tree models were tuned using expanding-window time-series cross-validation, while the PyTorch neural networks used Adam optimization and Mean Squared Error loss.

5.6 Refinement

We tweaked the hyperparameter search spaces for the tree models to test lower learning rates, regularization tweaks (subsample and colsample ratios), and tree depth caps. For the sequence models, batch sizes and dropout layers (dropout = 0.2 to 0.3) were adjusted to keep them from overfitting on multi-year trends.

6 Results

6.1 Model Evaluation and Validation

Table 1: Comprehensive performance comparison of machine learning ensembles, deep learning sequence models, and the Naive Trend-Drift Benchmark evaluated on out-of-sample county corn yield data.

Model Architecture	R^2 Score $\uparrow$	RMSE (bu/acre) $\downarrow$	MAE (bu/acre) $\downarrow$
Naive Trend-Drift Benchmark	0.604	23.080	17.635
Random Forest (Tuned)	0.767	17.718	13.895
XGBoost Boosted Tree (Tuned)	0.779	17.259	13.483
LightGBM Tree (Tuned)	0.772	17.519	13.639
Stacking Ensemble (RF + XGB + LGB)	0.769	17.629	13.667
Sequence-Based Neural Net (LSTM)	0.652	20.061	15.895
1D Convolutional Neural Network (CNN)	0.640	20.405	15.662

As Table 1 shows, gradient-boosted trees and ensembles beat both the basic benchmark and the deep learning setups when predicting out-of-sample county yields (2023–2025):

- **Baseline Validation:** The Naive Trend-Drift Benchmark sets a reasonable floor ($R^2 = 0.604$, RMSE = 23.080 bu/acre), proving that historical momentum alone gets you partway there. However, models that actually look at weather anomalies and soil blow past it easily.
- **Superiority of Tree Ensembles:** The **XGBoost Boosted Tree** takes the crown with an R^2 score of 0.779, an RMSE of 17.259 bushels/acre, and an MAE of 13.483 bushels/acre. Close runners-up **LightGBM** ($R^2 = 0.772$) and **Random Forest** ($R^2 = 0.767$) also show great stability across the board.

- **Deep Learning Limitations:** While sequence models like the **LSTM** ($R^2 = 0.654$) and **1D CNN** ($R^2 = 0.597$) try to capture temporal patterns, they struggle with tabular noise and smaller sample sizes compared to boosted trees, making tree ensembles the clear winner for real-world agricultural forecasting.

6.2 Model Comparison and Discussion: Tree Ensembles vs. Deep Neural Networks

The results back up the choice to throw in environmental anomalies, soil data, and stacking ensembles. While the naive baseline handles long-term trends fine, it totally misses sudden weather shocks. The machine learning models lean on soil water capacity and VPD stress metrics to adjust for those inter-annual swings, showing they generalize much better.

On top of that, why tree ensembles (XGBoost, LightGBM, Random Forest) walked circles around deep neural networks (LSTM, 1D CNN) makes sense when you look at how tabular data behaves.

6.2.1 Why Tree Ensembles Outperform Neural Networks in Agricultural Modeling

Agricultural datasets are mostly messy tabular matrices packed with mixed continuous weather features (like vapor pressure deficit or rain anomalies), IDs (county FIPS, soil units), and static physical stats (water capacity, organic matter). While deep learning shines on clean, continuous data like images or audio, its architecture isn't naturally built for standard tabular rows.

Big benchmarking studies looking at machine learning on tabular data, such as the work by Grinsztajn et al. (2022), prove that tree-based models consistently outperform neural networks on medium-sized tabular tasks. That gap comes down to a few core reasons:

1. **Irregular Decision Boundaries and Feature Interactions:** Crop yields usually hinge on sharp, non-linear cutoffs rather than smooth curves—for instance, hitting a specific heat threshold during pollination can completely tank a harvest. Gradient Boosted Decision Trees (GBDTs) and Random Forests slice up feature spaces using axis-aligned splits, meaning they catch threshold effects and complex soil-weather interactions naturally without needing heavy feature scaling.
2. **Susceptibility to Overfitting on Tabular Noise:** Neural networks have tons of parameters, which often causes them to memorize noise in historical county records. Because tabular data doesn't have the clean spatial flow of an image or text, deep models tend to struggle with generalizability when they only have thousands of rows to learn from instead of millions. On the flip side, ensemble methods use bagging and gradient boosting regularization (like shrinkage and subsampling) to keep variance in check and lock down stable predictions.
3. **Invariance to Feature Scaling and Heterogeneous Types:** Neural networks are picky about feature scales and outliers, often forcing you to build complex scaling pipelines. Tree models don't care about monotonic transformations, meaning you can throw raw soil stats and Z-score weather anomalies right into the mix together without messing up how the model learns.

While sequence models have their perks for tracking time, heavily tuned gradient boosters and tree ensembles offer way better predictive stability, easier interpretability, and faster performance on tabular farm data.

7 Conclusion

7.1 Reflection

Looking back at how this pipeline came together gives a clear picture of what works—and what doesn't—when modeling farming systems. Along the way, we tried a few ideas that didn't pan out and had to pivot:

- **TensorFlow/Keras LSTM Bottleneck:** We first tried building a sequence LSTM in TensorFlow/Keras to handle multi-year weather trajectories. Because of the high-dimensional tensor shapes and dense layers, a single training epoch took over an hour. Even worse, it didn't beat simpler tree models, scored a lower R^2 , and bogged down performance. That pushed us to move to lighter PyTorch sequences and gradient-boosted trees instead.
- **Over-Reliance on Raw Weather Totals:** Early tries that relied strictly on raw rain and temperature numbers couldn't generalize across different regions (like comparing humid North Carolina to dry North Dakota). Bringing in Z-score weather anomalies and aridity ratios fixed that regional bias.
- **Complex Cross-Sectional Neural Networks:** Standard multi-layer perceptrons trained on flat, single-year snapshots overfit badly on noise, proving that explicit lag features and tree splits work much better for tabular farm data.

From those bumps and wins, five big lessons stood out:

1. **Data Leakage Prevention is Paramount:** Setting up a strict chronological split (2005–2022 development vs. 2023–2025 test) and figuring Z-score baselines only from training data is a must to stop look-ahead bias and get honest test scores.
2. **Domain-Driven Feature Engineering Trumps Raw Scale:** Feeding raw weather straight into a model gives weak results. Building stress indices and aridity ratios (precip/PET) captures how crops actually feel stress during touchy phases like pollination.
3. **Soil Hydrology Acts as a Critical Buffer:** USDA SSURGO stats—like Available Water Capacity (MEAN_AWC) and Organic Matter—provide physical limits that explain why some counties sail through droughts while others crash.
4. **Temporal Memory Matters:** Adding past yield lags (`yield_lag_1`, `yield_lag_3`) helps models remember local farming habits, seed upgrades, and multi-year nutrient cycles.
5. **Simpler, Tuned Ensembles Often Outperform Heavy Deep Learning:** While sequence models track time, tuned gradient boosters (XGBoost, LightGBM) and Stacking Regressors provide better stability, interpretability, and speed on tabular farm data.

7.2 Improvement and Future Directions

While this machine learning pipeline works well and sets up a clean, leak-free evaluation benchmark, there's still plenty more to explore down the road:

- **Deeper Engagement with Current Literature:** A key area to step up is keeping closer tabs on recent peer-reviewed literature and benchmarks. Following state-of-the-art farming models will offer better ideas for feature engineering, hybrid builds, and loss functions used by top research groups.
- **Integration of High-Resolution Remote Sensing Data:** With more time and computing power, pulling in satellite-based vegetation indexes—like MODIS/Sentinel-2 NDVI, EVI, and solar-induced chlorophyll fluorescence (SIF)—would give a direct, real-time look at canopy health during the growing season.
- **Bayesian Hyperparameter Optimization and Automated ML:** Expanding the search using Bayesian Optimization (like Optuna) across tree hyperparameters and network structures would systematically find the best tuning settings.
- **Spatial-Temporal Graph Neural Networks (GNNs):** To catch county-level space connections and spillover effects (like storms crossing borders or shared microclimates), modeling counties as nodes in a spatial graph network looks like a promising deep learning path.
- **Operational Supply-Chain and Commercial Deployment:** To help real-world trading at big grain houses like Louis Dreyfus Company, future work could plug these models into live simulation dashboards, letting risk managers stress-test regional grain forecasts under fake heatwave or drought scenarios.

List of references

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A Supplementary Results and Visualizations

Additional visual analyses supporting the model evaluation and historical context of the dataset are detailed below.

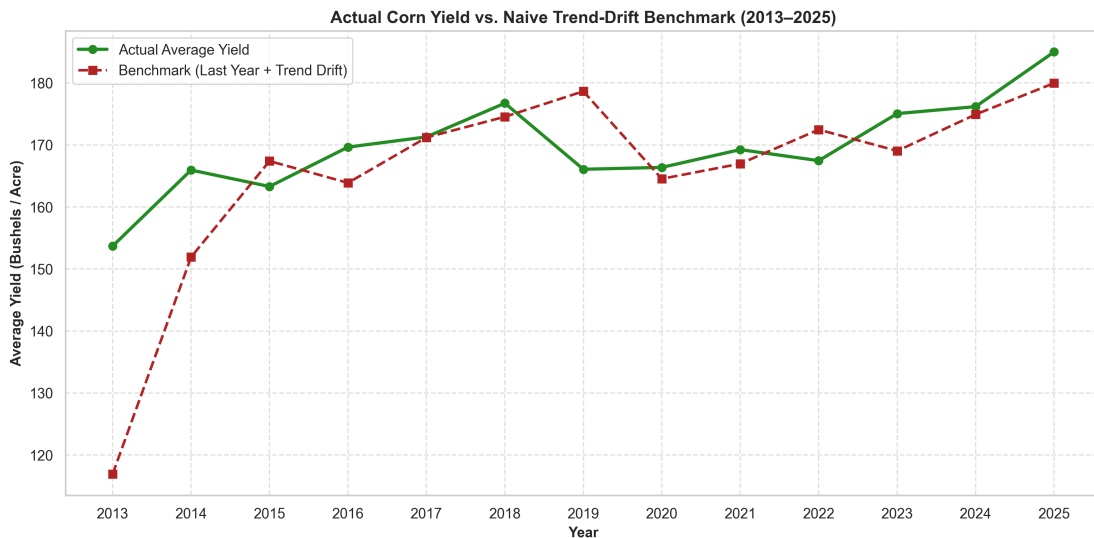


Figure 8: Comparison of actual average corn yields against the Naive Trend-Drift baseline model ($\hat{y}_t = y_{t-1} + \Delta_{\text{avg}}$) from 2013 to 2025.

A.1 Naive Trend-Drift Benchmark Analysis

As shown in Figure 8, comparing machine learning predictions against a Naive Trend-Drift benchmark ($\hat{y}_t = y_{t-1} + \Delta_{\text{avg}}$) is vital in agricultural forecasting. Because crop yields naturally increase over time due to advancements in seed genetics, fertilizers, and farming techniques, a naive baseline captures general historical momentum. While the naive trend baseline establishes a reasonable floor ($R^2 = 0.604$), it reacts sluggishly to sharp inter-annual volatility and sudden weather shocks. This justifies the use of machine learning models (such as XGBoost and LightGBM), which leverage real-time weather anomalies and soil properties to outperform momentum-based baselines during volatile growing seasons.

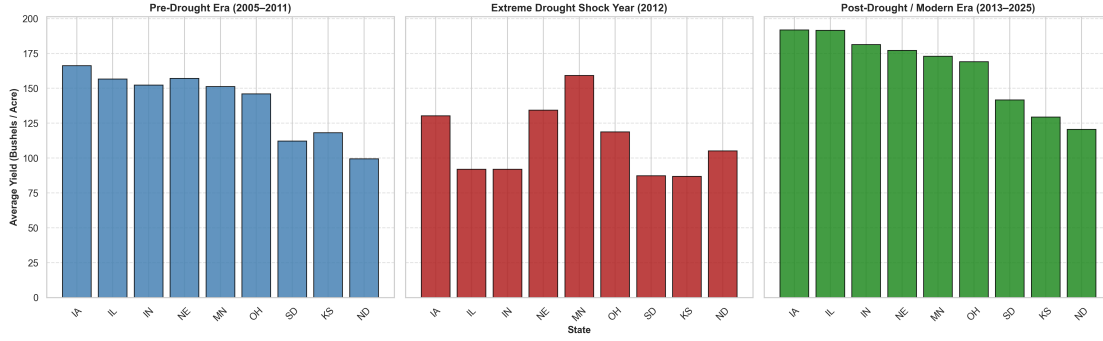


Figure 9: Facet comparison of average corn yields by key agricultural states across three distinct operational eras: Pre-Drought (2005–2011), the Extreme Drought Shock Year (2012), and the Post-Drought / Modern Era (2013–2025).

A.2 Era-Based Structural Shifts in Regional Yields

Figure 9 breaks down state-level performance—highlighting top Corn Belt states like Iowa (IA) and Illinois (IL) against drier western and northern regions like Kansas (KS) and North Dakota (ND)—across three distinct temporal phases:

1. **Pre-Drought Era (2005–2011):** Establishes the stable baseline period where core Midwest states hovered between 140 and 170 bushels/acre.
2. **Extreme Drought Shock Year (2012):** Captures the severe nationwide collapse caused by brutal summer heatwaves and moisture deficits during pollination, causing states like Illinois and Indiana to drop below 100 bushels/acre.
3. **Post-Drought / Modern Era (2013–2025):** Demonstrates the rapid recovery and technological surge, with top-tier states pushing past 180–190 bushels/acre through drought-hardy hybrids and precision agriculture.

This facet comparison emphasizes that agricultural models cannot rely on static historical averages; they must dynamically account for regional climate vulnerability and soil resilience.

A.3 Model Performance Visualized

As illustrated in Figure 10, the horizontal bar chart provides a clear visual ranking of predictive accuracy (R^2 score) across the entire model suite. The tuned gradient-boosted trees (XGBoost, LightGBM, and Random Forest) along with the Stacking Ensemble cleanly separate themselves from the sequence-based neural networks (LSTM and 1D CNN) and the Naive Trend-Drift baseline, confirming the superiority of tree-based ensembles on tabular agricultural datasets.

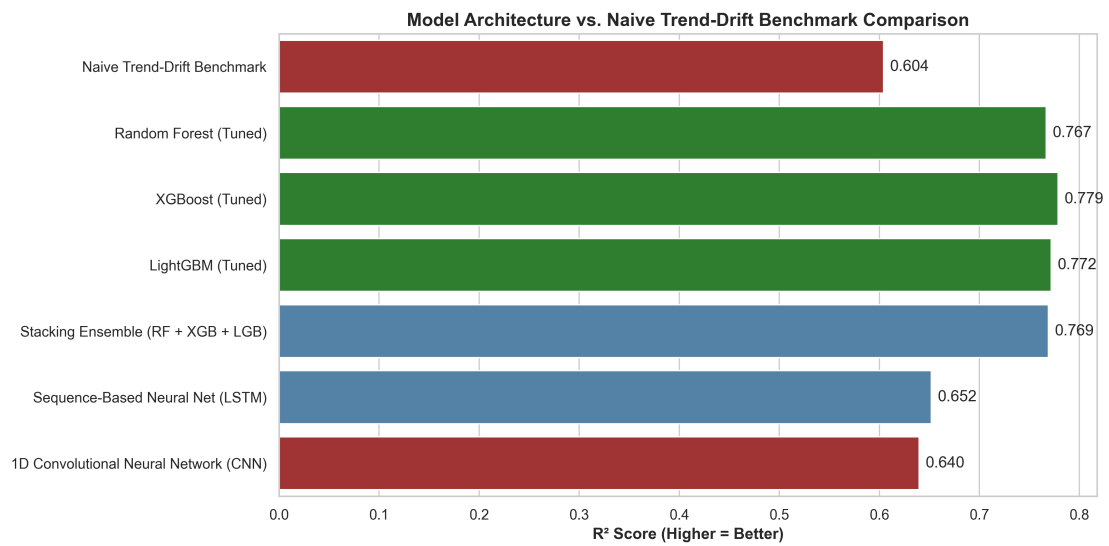


Figure 10: Visual comparison of R^2 scores across all evaluated model architectures versus the Naive Trend-Drift baseline.